

Hot In-place Recycling (HIR)

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Methods of Asphalt Recycling



Hot In-place Recycling

- Hot Mix Asphalt Recycling (Central Plant)
- Cold Mix Asphalt Recycling (Central Plant)
- Cold In-place Recycling
- Full Depth Reclamation

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Hot In Place Recycling Process

- On site, hot in-place method that rehabilitates deteriorated asphalt pavements and thereby minimises the use of new materials
- Depth recycled: 20-60 mm (25 mm most common) (1 to 2.4 inches)
- Therefore deep ruts or deep problems cannot be treated by this recycling option.

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Pavement Distresses Treated by HIR

- Raveling
- Potholes
- Bleeding
- Skid resistance (Friction number)
- Shallow Rutting

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Pavement Distresses Treated by HIR (Continued)



- Corrugations
- Shoving
- Slippage, longitudinal, transverse, and reflection cracking
- Poor ride quality caused by swells, bumps, sags, and depressions

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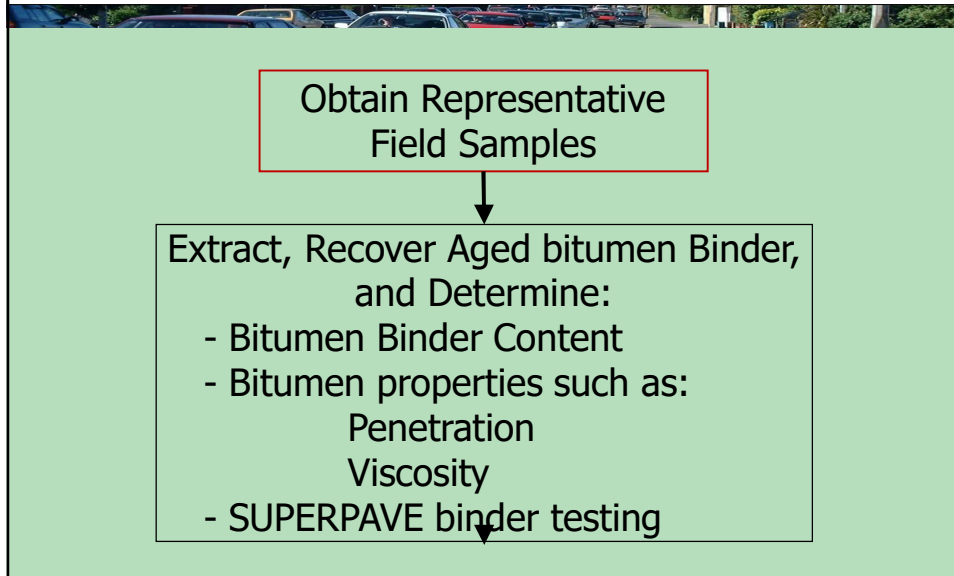
HIR Mix Design Process



- Evaluation of the existing HMA including asphalt binder, aggregates, and mix properties
- Determining whether the existing asphalt binder needs rejuvenation
- Selecting the type and amount of recycling agent

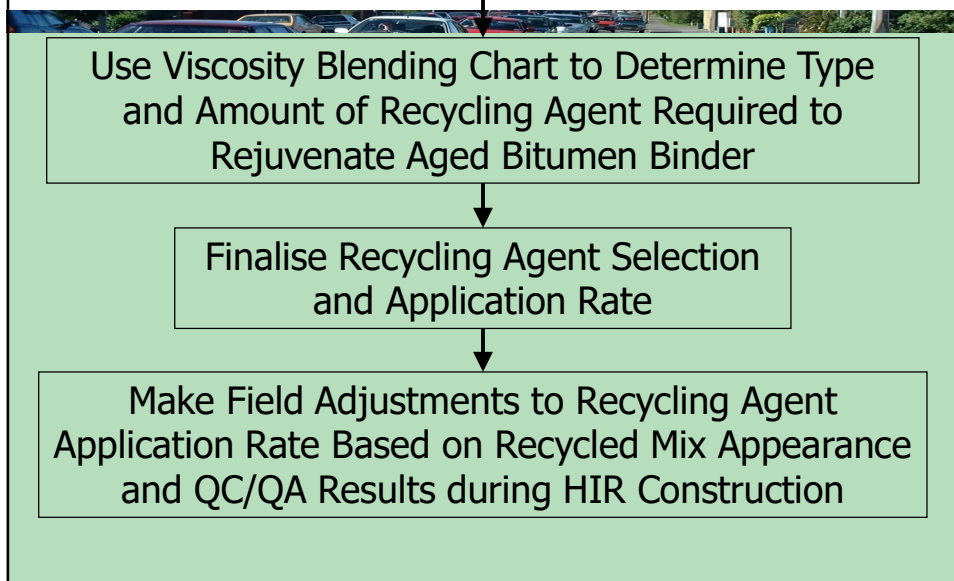
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Procedure for HIR Mix Design



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Blending Chart HIR Mix Design



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HIR Mix Design Process

- Determining the amount of admix including aggregate gradation, type, and amount of soft, new bitumen binder
- Preparing and testing both bitumen and mix specimens in the laboratory
- Evaluating test results and determining the optimum combination of admix and recycling agent

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HIR Laboratory Mix Design

Evaluation of the Existing Pavement

- Bitumen binder content
- Rheological properties of the Bitumen
- Gradation and aggregate properties

Characterisation of the Virgin Materials

- Gradation and aggregate properties
- Type and amount of new asphalt binder
- Type and amount of recycling agent.

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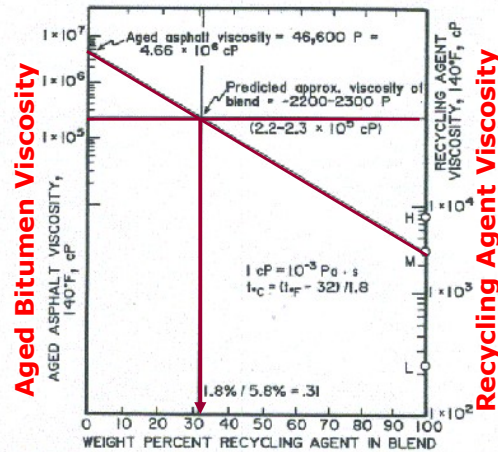


Figure 7-5. Nomograph for Viscosity for Design of Recycled Mixtures (ASTM D4887)

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- R,T, V log viscosities in Pa.s at a single temperature.
- R=Log viscosity of the rejuvenating agent
- T= log target viscosity
- V= Log viscosity of the bitumen extracted from RAP

$$r = \frac{\text{Log}(V+3) - \text{Log}(T+3)}{\text{Log}(V+3) - \text{Log}(R+3)}$$

$$r = \frac{\text{mass of rejuvenator}}{\text{mass of rejuvenator} + \text{mass of binder in RAP}}$$

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Characterisation of the Combined Recycled Mix

- Rheological properties of the asphalt binder
- Gradation and aggregate properties
- Recycled mix void properties and physical properties.

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Basic Steps for HIR in the Field

- Soften pavement with heat
- Scarification or mechanical removal of softened material
- Mixing with recycling agent, new aggregate, new binder, or new mix
- Laydown and paving

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Methods of Hot In-place Recycling

Methods and Equipment

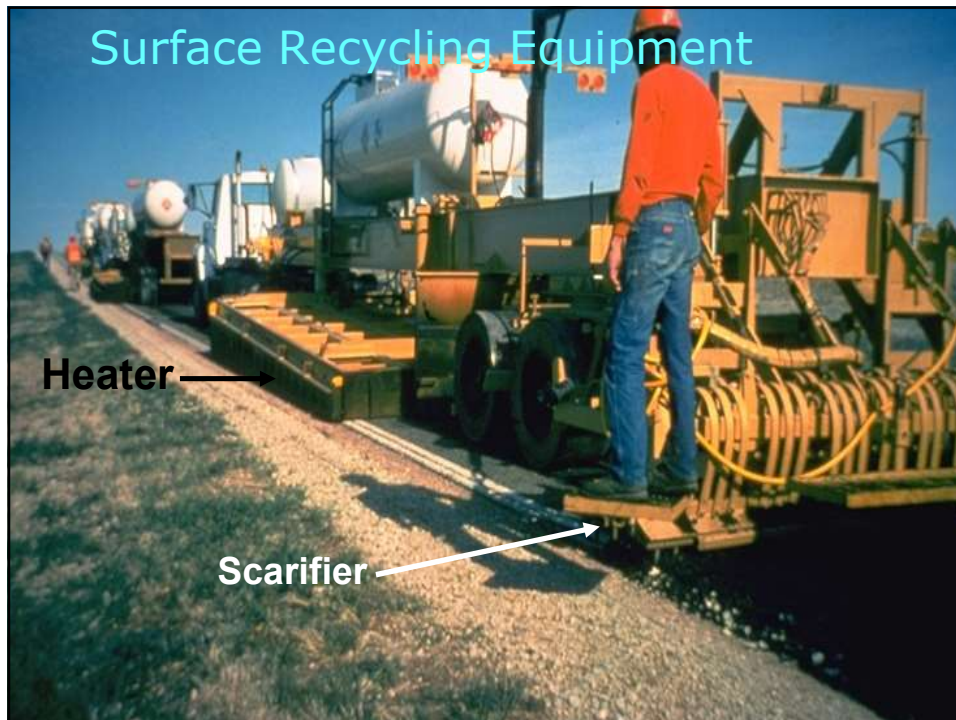
- I. Surface Recycling
- II. Repaving
- III. Remixing

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Surface Recycling

- Rehabilitation process that restores cracked, brittle, and irregular pavement in preparation for a final thin wearing course
- Depth of 25 mm (1 inch) most common, 40 mm (1 1/2 inches possible)

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Surface Recycling Methods



- Single-Pass Method
- Two-Pass Method (HMA overlay placed as a separate operation after surface recycling)

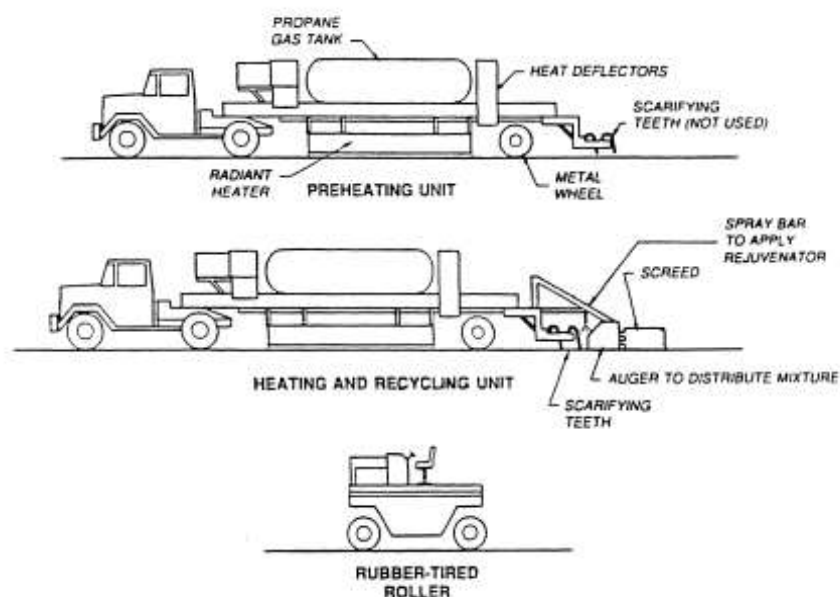
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Surface Recycling

- Radiant or infrared heating
- Propane - most common fuel
- Spring loaded scarifiers
- Effectiveness of adding recycling agent without mixing

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Schematic of Surface Recycling



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Remixing Process



- Heating asphalt pavement 37.5-60 mm (1 1/2 to 2.4 inches)
- Scarification and collection of RAP in windrows
- Add virgin aggregate, recycling agent, or virgin HMA mix
- Mix in a pugmill
- Spread recycled mix and compact

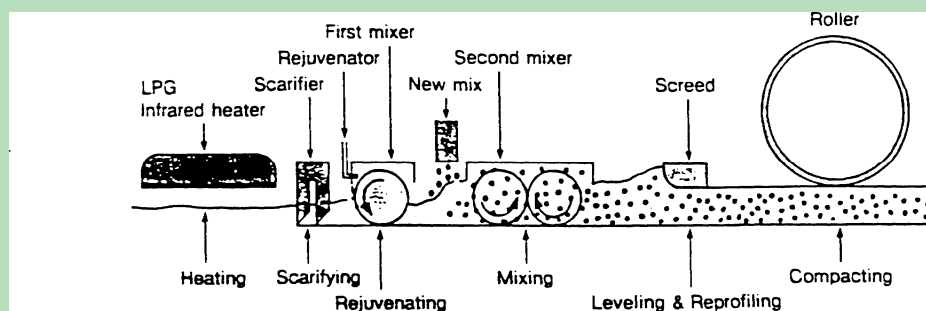
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Advantages of Remixing

- Elimination of rutting, cracking, and oxidation in top 50 mm (2 inches) of asphalt pavement
- Restoration of existing asphalt mix to desired mix composition and strength by adding new aggregate, binder, or new mix

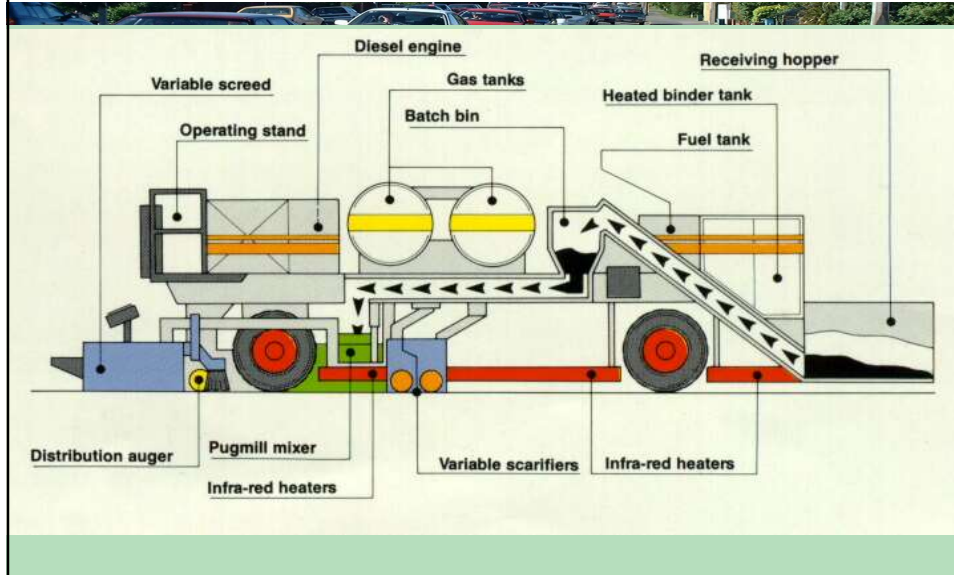
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Schematic of Remixing Method



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Single Pass Remixing Equipment



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Hot In-place Recycling



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Repaving



- Definition: surface recycling method combined with simultaneous overlay of new HMA to form a thermal bond between recycled and new layers
- Depth: 25 to 50 mm (1 to 2 inches)

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Steps in Repaving Process

- Multiple heating units to soften asphalt surface
- Scarify softened pavement
- Add and mix recycling agent with RAP

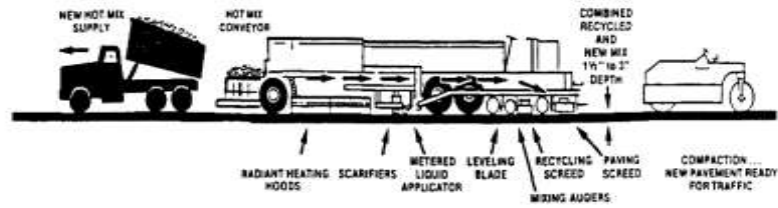
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Steps in Repaving Process

- Spread the recycled mix with screed
- Spread the new mix with screed on top of the recycled mix
- Compact both layers of recycled and new mixes

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Schematic of Single - Pass Method



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Pavement After Repaving



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Problem with Crack Filling Material



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Problem with Crack Filling Material



- Flare-ups can occur under preheaters
- Can use strip of sand or hydrated lime
- If present in excessive amount, material must be removed

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New Equipment Advantage

- Old uses high intensity infrared heaters
 - Overheat and damage asphalt
 - May cause emission
- Currently using indirect radiant or infrared
- New equipment uses hot air/low level infrared heat

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Quality Control (QC) / Quality Assurance (QA)

Item	Method
Depth of scarification	Measure depth
Application rate of recycling agent	Calculate from quantity used Determine bitumen content before and after adding recycling agent

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QC/QA (Continued)

Item	Method
New mixture addition rate	Calculate from quantity used
Temperature of mix before compaction	Measure
Properties of asphalt binder in recycled mix	Recover bitumen by extraction methods and test

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Advantages of HIR

- Pavement geometrics preserved
- Surface cracks can be eliminated
- Ruts, shoves, bumps can be corrected
- Aggregate gradation and bitumen content can be modified
- Relatively cheap, minimal hauling costs

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HIR Disadvantages/Problems

- Does not work well with pavements with too much variation in HMA mix composition.
- Presence of any fabrics or interlayers in the treatment depth over 25% cannot be recycled
- Deep ruts greater than $\frac{1}{2}$ of the anticipated HIR treatment depth imposes limitations on the type of HIR process

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HIR Disadvantages/Problems

- Large stone mixes may not be recycled
- Presence of rubber in the mix cause difficulties during construction
- Moisture content in the existing pavement reduces the production rate of the HIR process.

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Savings with HIR

Agency	HIR Process	Saving Over Control (%)
Mississippi SHD	Remixing	40
Oregon DOT	Remixing	17
Texas DOT	Remixing	34

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Savings with HIR

Agency	IR Process	Savings Over Control (%)
Florida DOT	Repaving	25
FAA Texarkana, Texas	Repaving	50

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HIR Production

- Production with HIR train
 - 2 to 4 lane kilometers
(1 ¼ to 2 ½ lane miles per day)

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HIR Performance

■ Performance

- HIR Surface Recycling without surface treatment 2 – 4 years
- HIR Surface Recycling with surface treatment 6 – 10 years

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Production and Performance (Continued)

■ Performance

- HIR Remixing 7 – 14 years
- HIR Remixing with HMA overlay 7 – 15* years
- HIR Repaving 6 – 15* years
- * Equivalent to agency's thick lift HMA service life.

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